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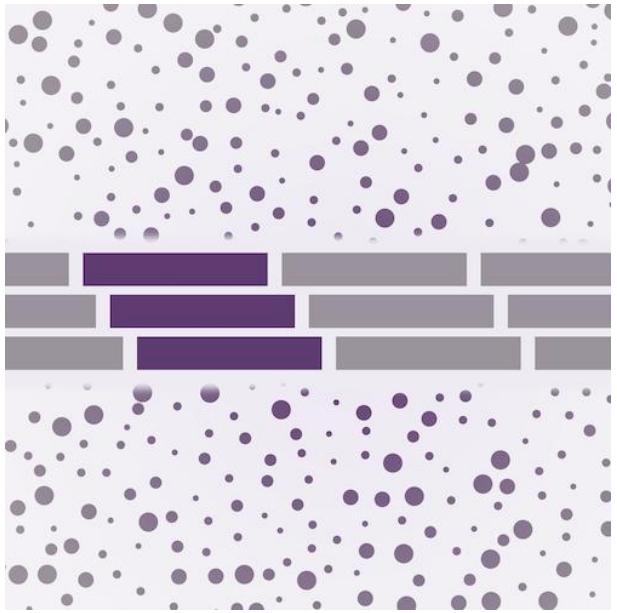
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Canadian Bioinformatics Workshops



Module #1b: Application Scenarios

Dr. Erin Pleasance

CBW High-Throughput Genomics

November 4-6, 2026

CANADA'S MICHAEL SMITH GENOME SCIENCES CENTRE

Scenarios

- We will go through a few example scenarios and discuss what approaches might be relevant
- You will be discussing in groups of 3-4 people
- The slide deck we go through will be posted at the end of today's workshop

Introductions (10 mins)

- Find a group of 3-4 people
- Introduce yourself and your research background/interests
- Share with your group: What is something you learned or found interesting so far today?

Scenario 1: Pneumonia outbreak

- Several hospitals have reported patients admitted with severe pneumonia
- We have samples from their respiratory tract
- We want to know what is causing the pneumonia and whether these outbreaks are related



Wikipedia.org

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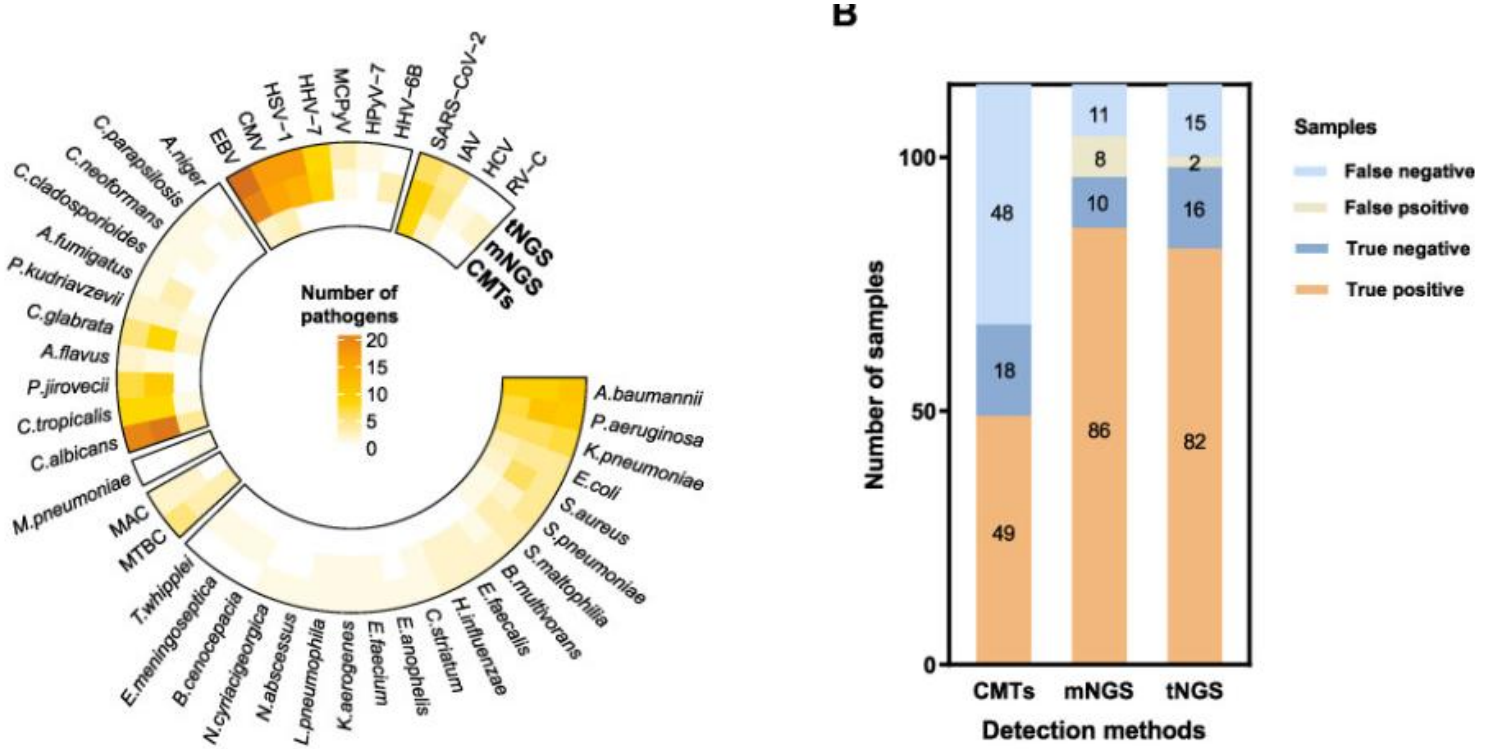
Wikipedia.org

Discuss:

- What kind of sequencing approaches might be useful?
- What would be the benefits or limitations of these approaches?

Scenario 1: Pneumonia outbreak

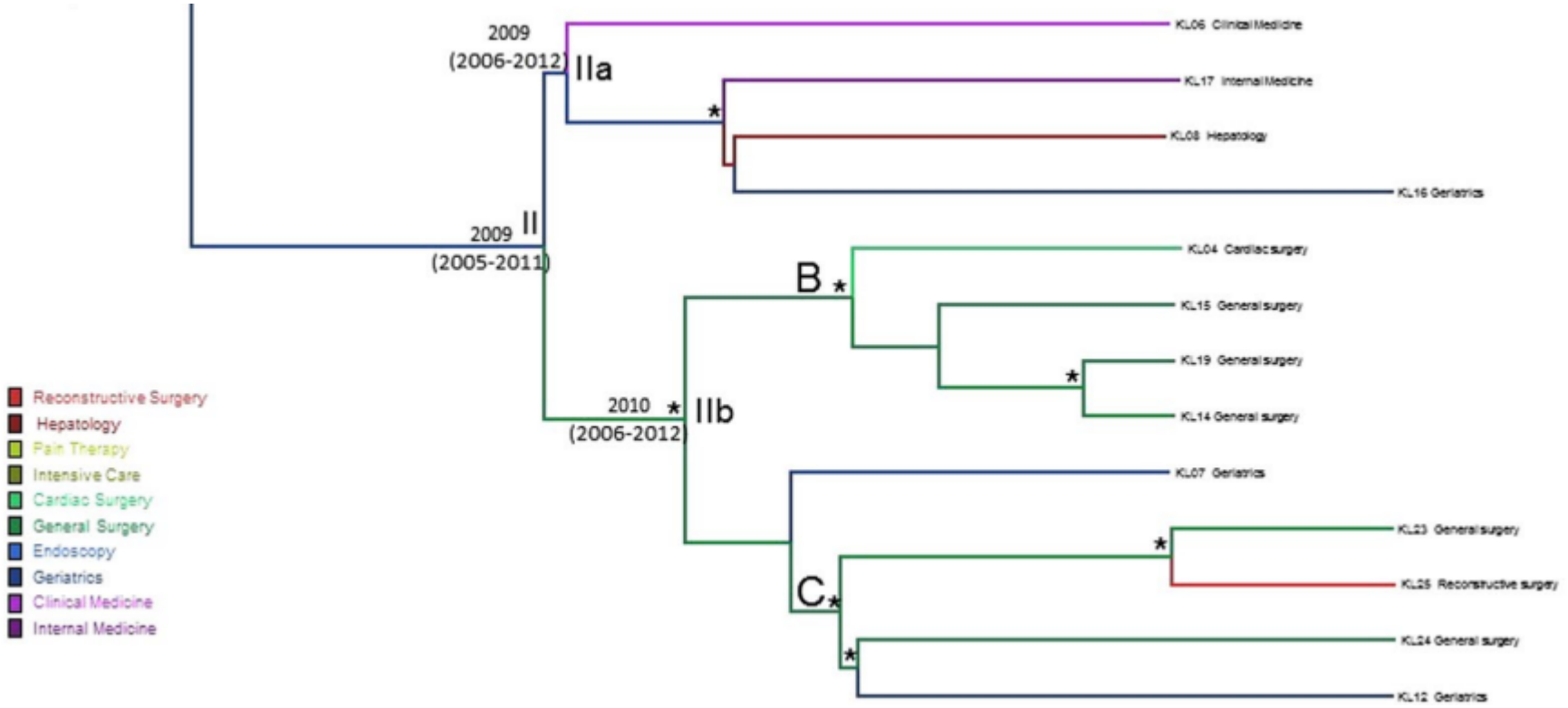
- Targeted vs. metagenomic: Cost, complexity, comprehensiveness



PMID: 39872133

Scenario 1: Pneumonia outbreak

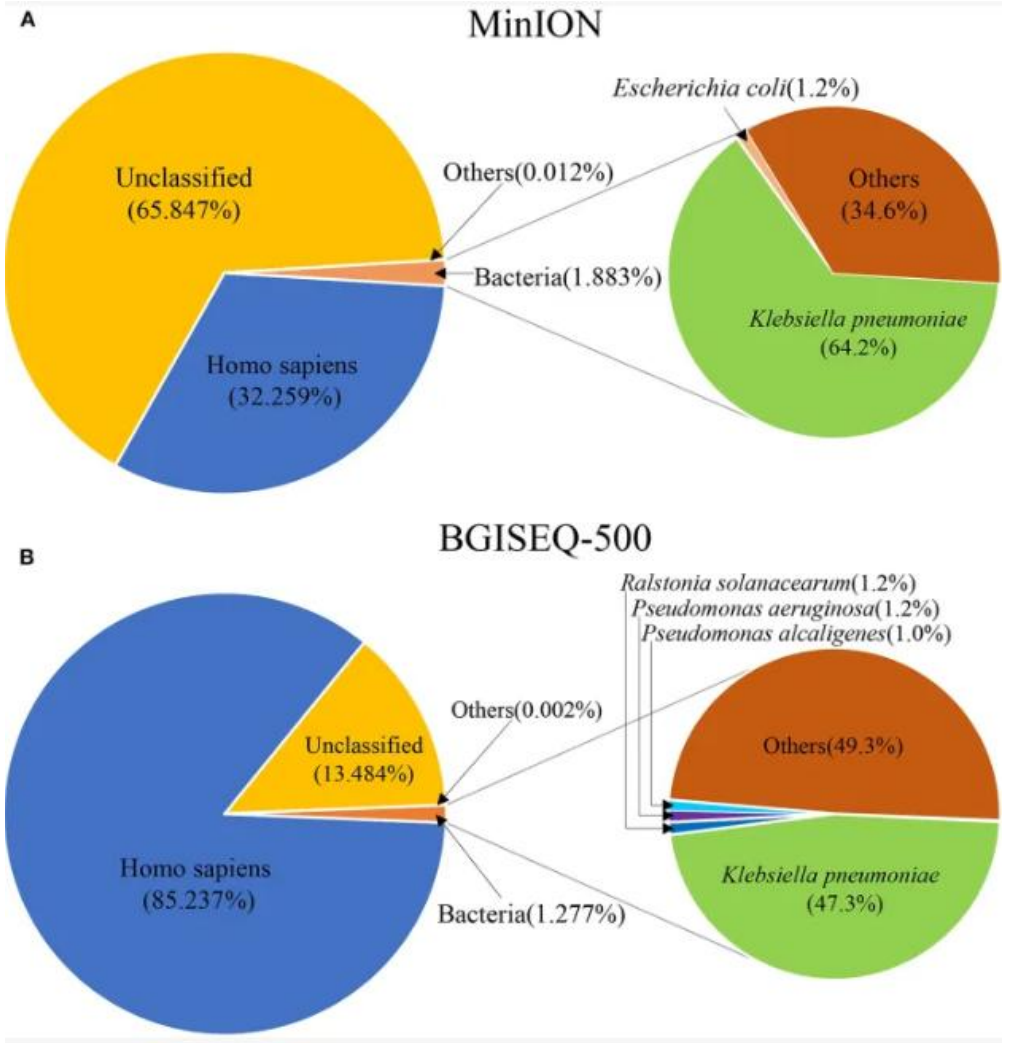
- Targeted vs. metagenomic: Phylogeny and drug resistance



PMID: 28615687

Scenario 1: Pneumonia outbreak

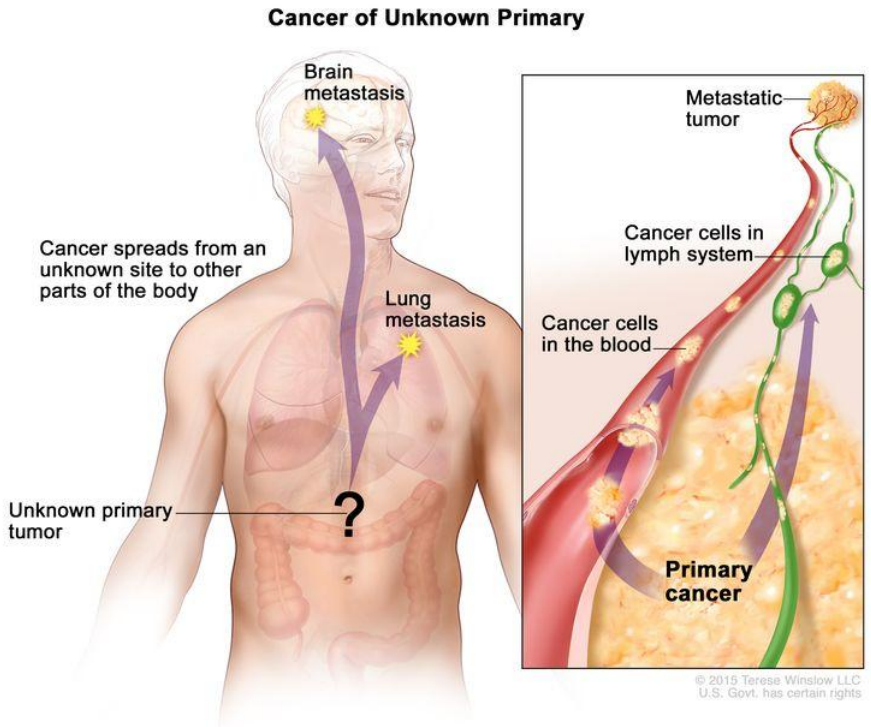
- Long vs short read: Time and accuracy
  - “It took ~1 min from beginning of sequencing to obtain the first specific reads of K. pneumoniae by MinION ... resistance genes including blaSHV-12, aac(3)-IIa and blaKPC-2 were detected within one and a half hours"



PMID: 32432051

Scenario 2: Unknown cancer

- A patient has a cancer of unknown origin with unclear pathology
- We have a sample of the tumour from the lung and of the patient's normal blood
- We want to know what type of cancer it is and how it might be treated



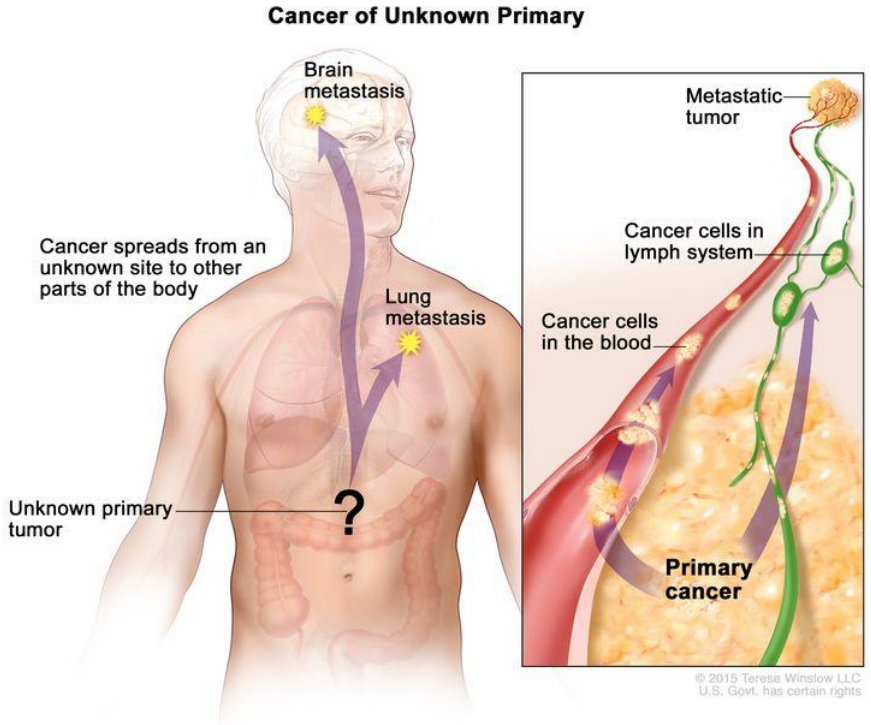
www.cancer.gov/types/unknown-primary/patient/unknown-primary-treatment-pdq

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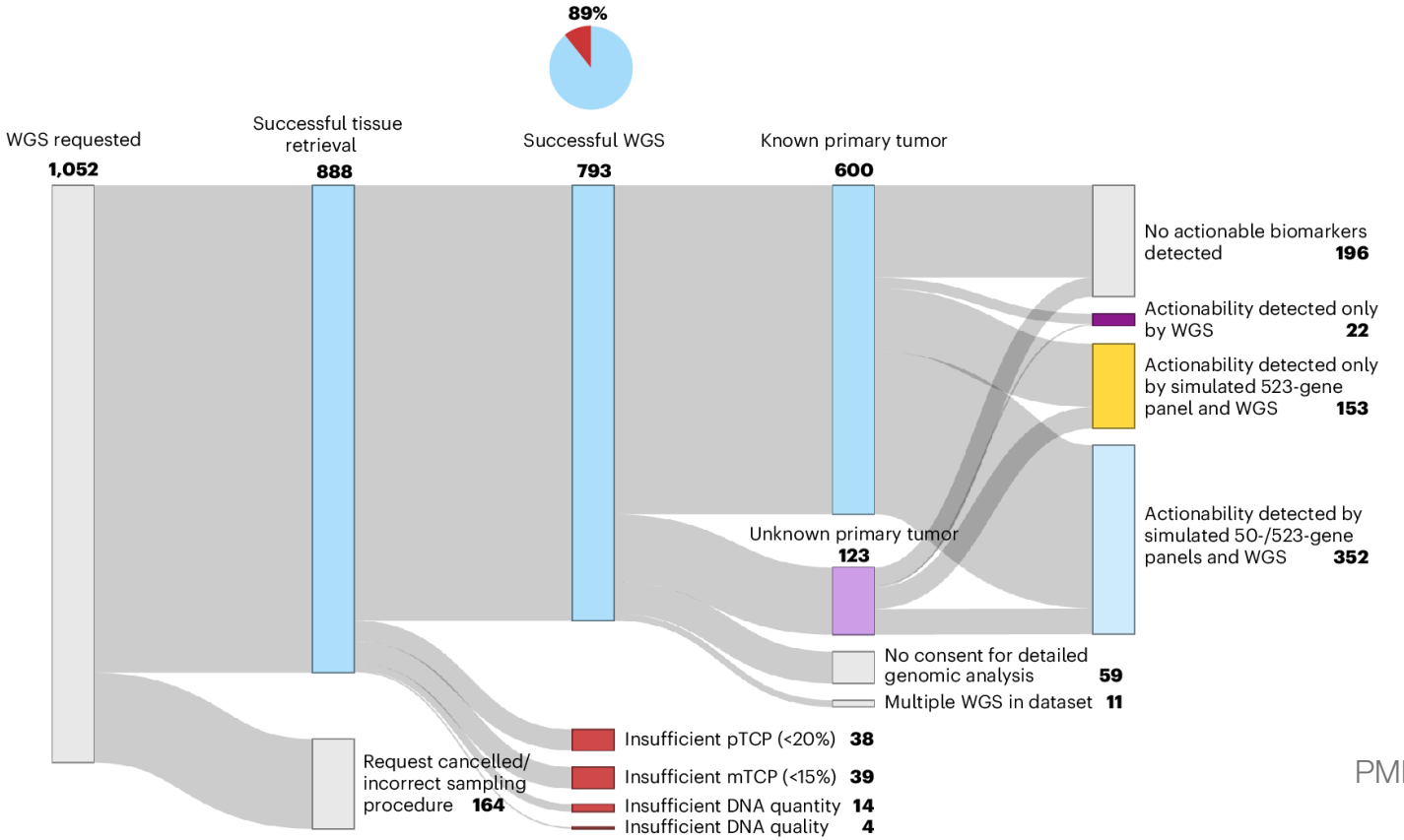
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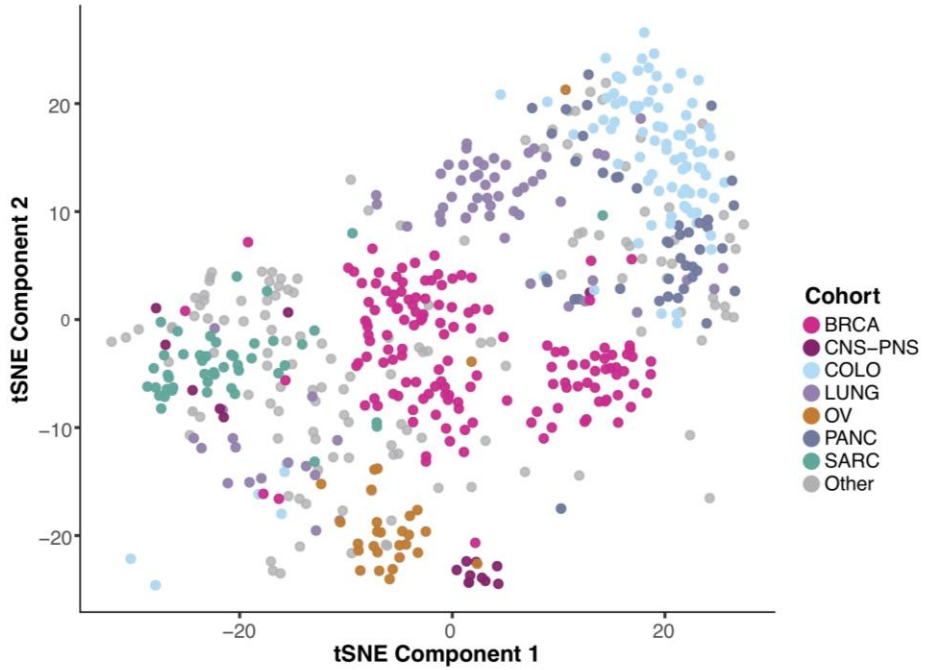
- Targeted vs. whole genome: Cost, complexity, comprehensiveness



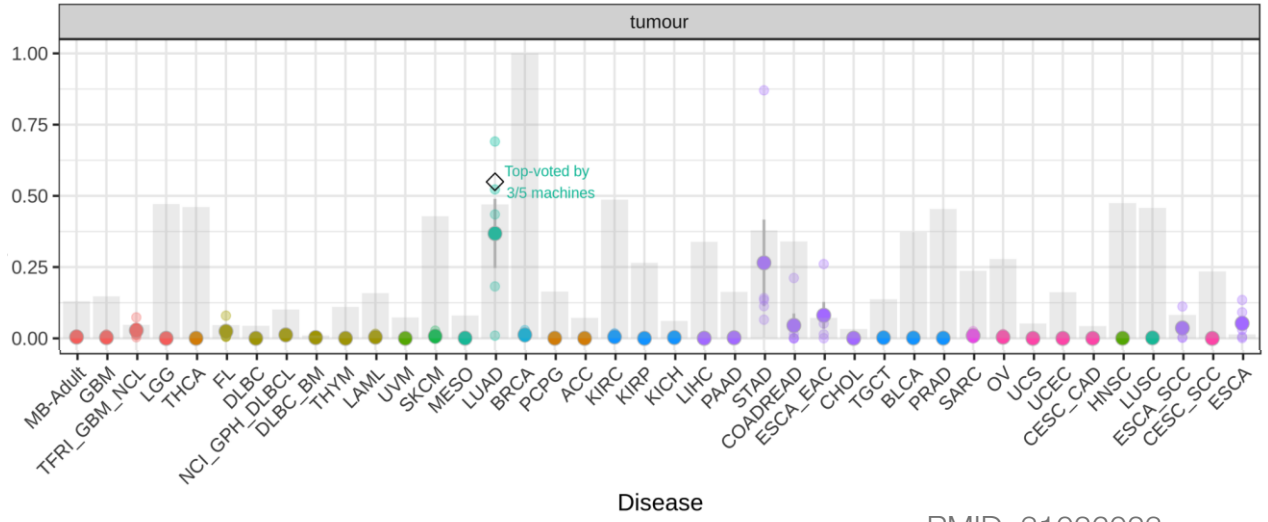
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Scenario 2: Unknown cancer

- RNA-Seq to aid with diagnosis



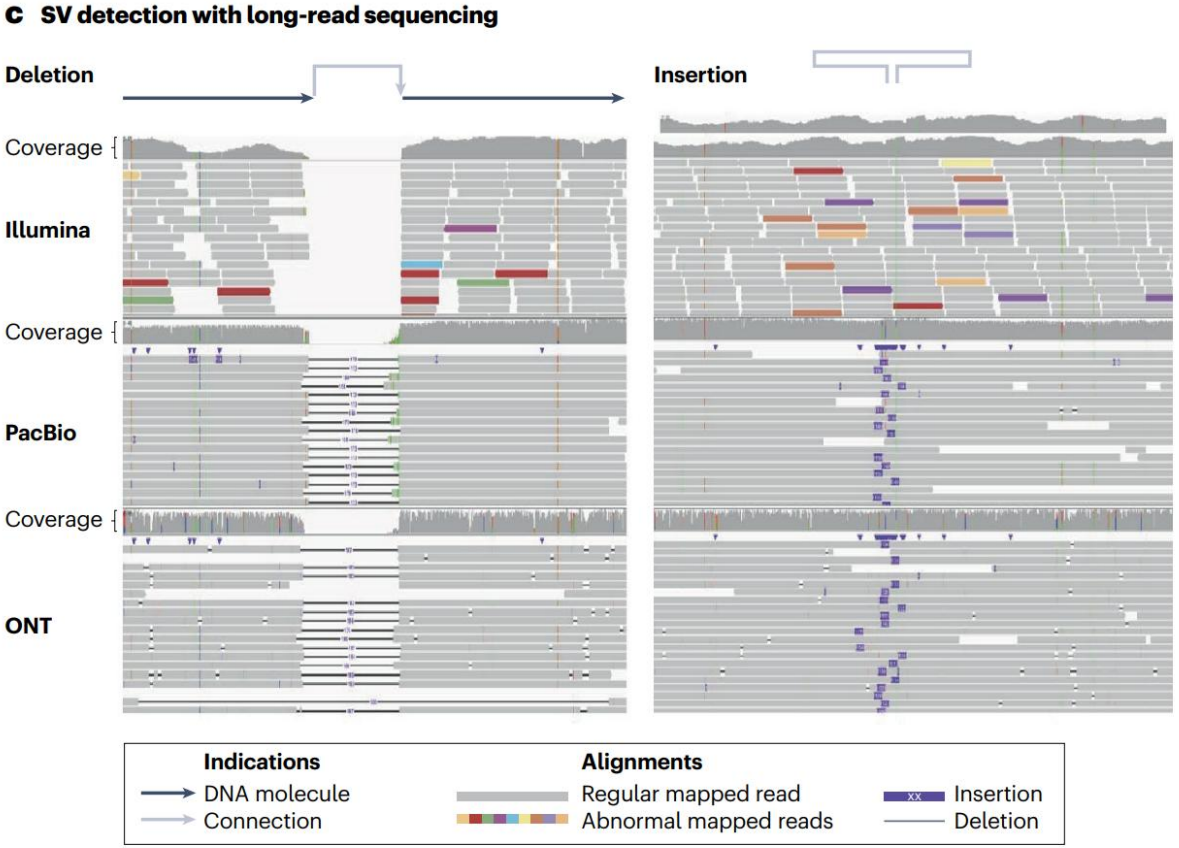
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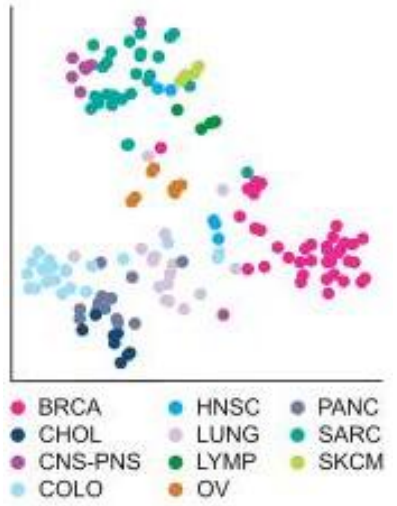
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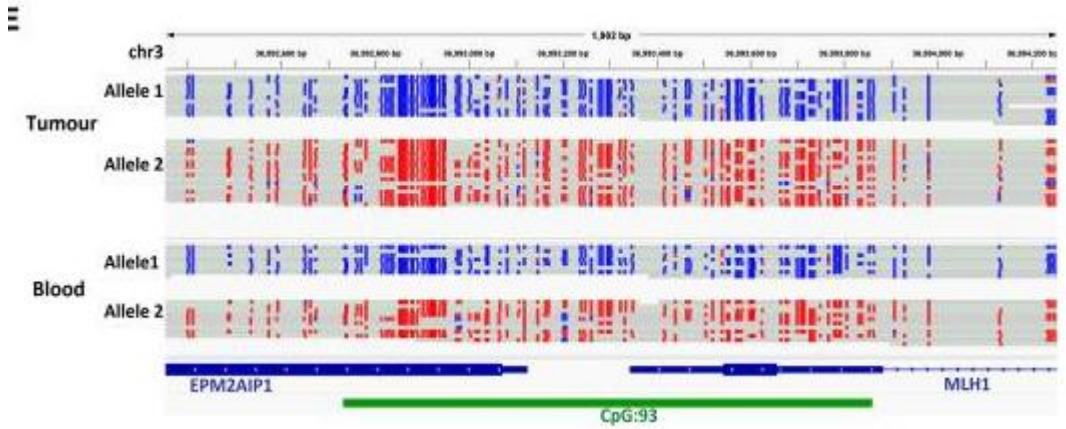
Scenario 2: Unknown cancer

- Long read vs short read: SV detection and methylation



PMID: 40817395





PMID: 39406235

Scenario 3: Novel species

- A rare new species has been discovered, the jackalope
- We have obtained tissue and blood samples
- We want to assemble a reference genome and understand what this species is related to



Harold Sanborn

Scenario 3: Novel species

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Discuss:

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Harold Sanborn

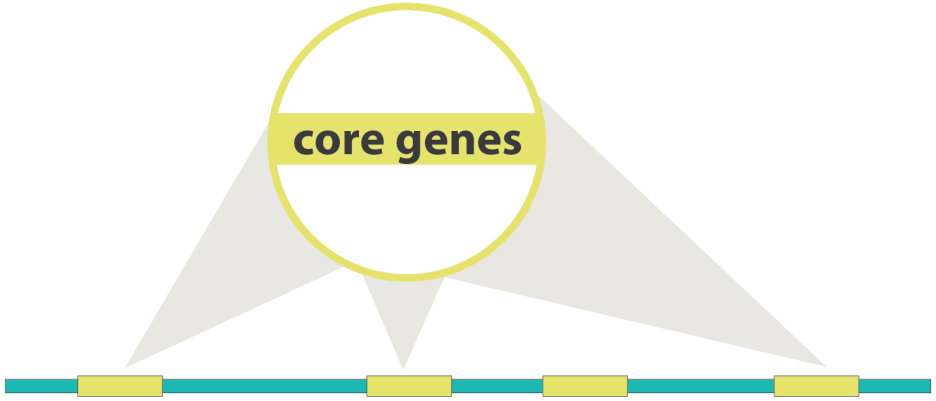
Scenario 3: Novel species

- Assembly quality

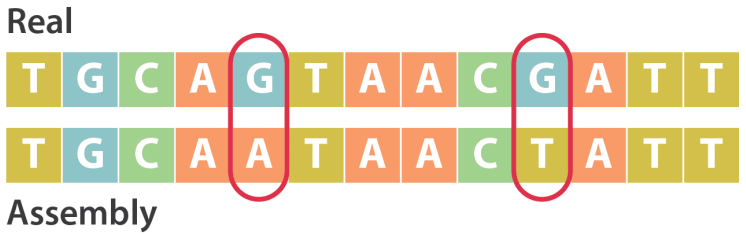
Contiguity



Completeness



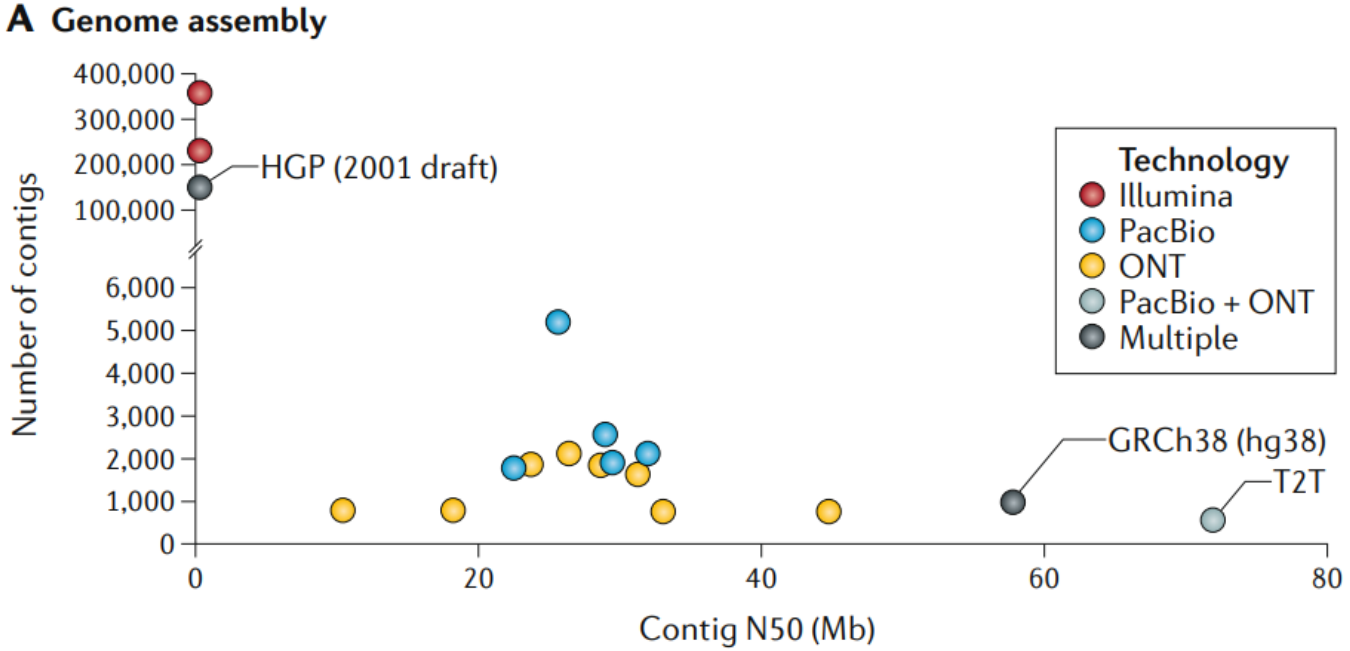
Correctness



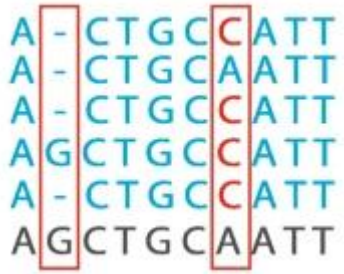
https://training.galaxyproject.org/training-material/topics/assembly/tutorials/assembly-quality-control/slides-plain.html

Scenario 3: Novel species

- Assembly quality: Long reads and short reads are complementary



PMID: 32504078



PMID: 34775997

Your own research

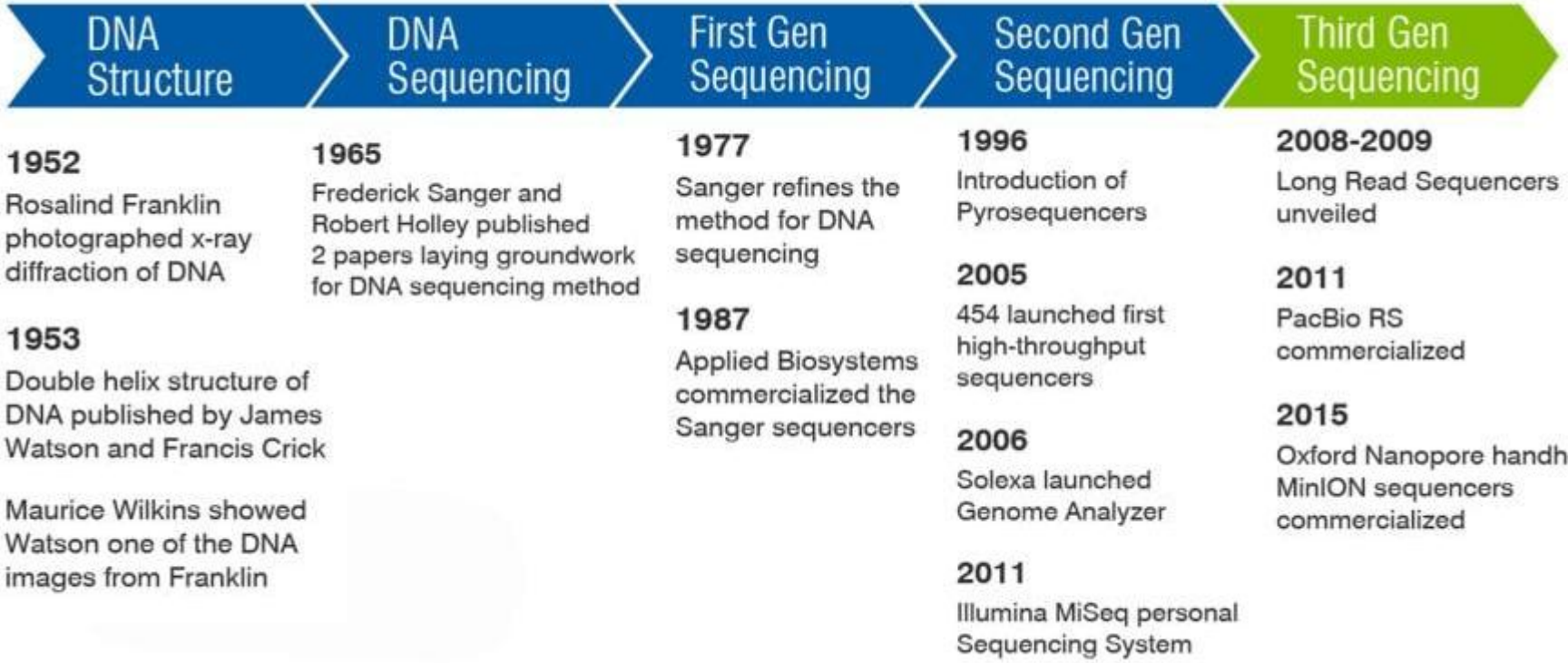
- What is a research question you have? Explain it briefly to your group.
- What technologies do you think might best apply to this question and why?
- What would be some of the limitations or challenges of using these technologies for your application?

Summary

Illumina sequencing	• Widely used with mature protocols and analysis tools • Highest throughput and lowest cost per base • Extremely accurate for SNVs and small indels • Ideal for WGS, RNA-Seq, single-cell seq, large cohorts	• Short reads, typically paired 150 bp • Structural variants, repeats, phasing can be challenging • Requires additional prep to detect epigenetic changes
PacBio SMRT sequencing	• Long reads, typically 10-25kb • Very high accuracy, with error rates less than 0.1% • Native methylation detection • Excellent for de novo assembly and structural variants	• High cost per base, making it the most expensive option for most applications • Lower throughput compared to other technologies • Requires high quality, molecular weight DNA
Oxford Nanopore	• Very long reads, 10K-100K+ bp • Direct detection of DNA & RNA epigenetic modifications • Portable and flexible, allowing for sequencing in remote locations or in real-time	• Somewhat higher cost and lower throughput than short read sequencing • Higher variability in accuracy compared to other technologies, particularly in homopolymers

Adapted from https://pmc.ncbi.nlm.nih.gov/articles/PMC10703097/table/t1/

DNA sequencing history



Newer technologies competing in short-read space:

2022
AVITI (Element Biosciences)

2023
Complete Genomics

2024
Ultima

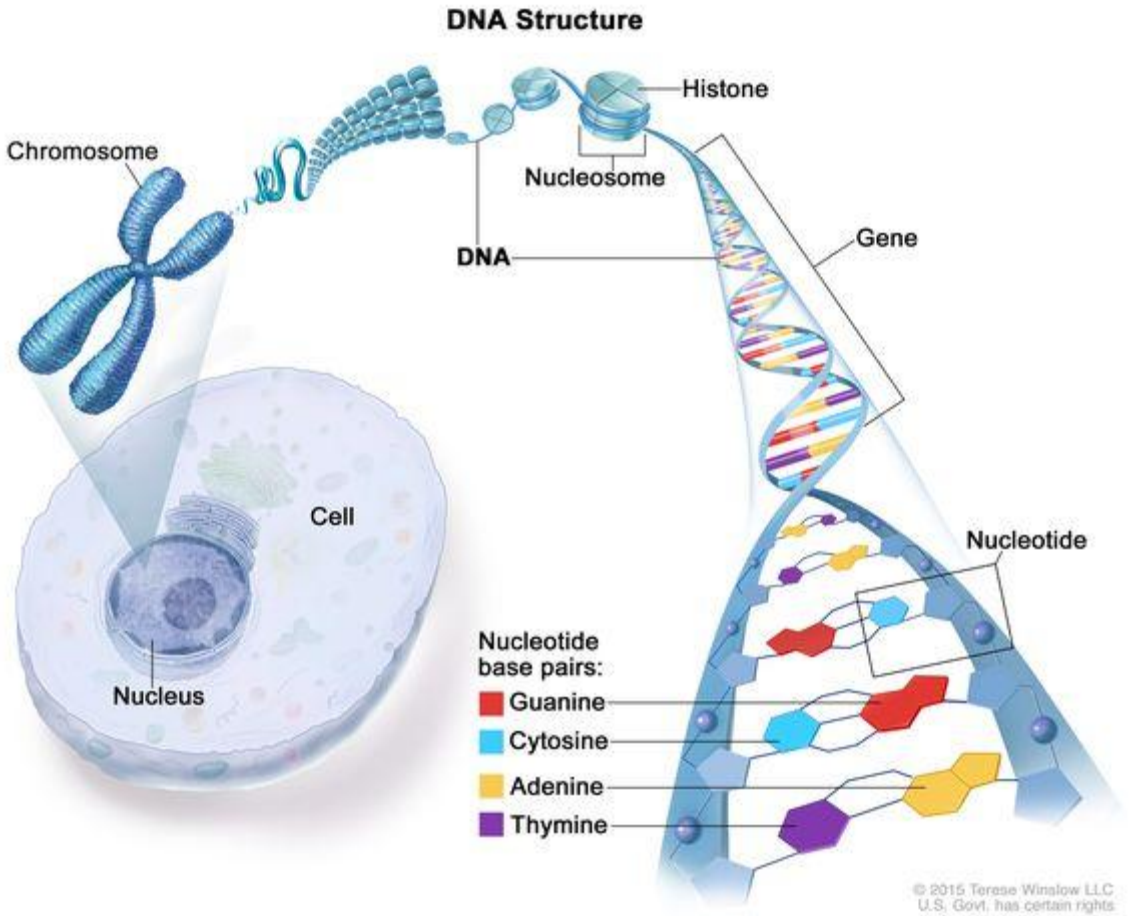
https://www.criver.com/eureka/history-dna-sequencing

Summary

- Technology is constantly changing
- Enables more science:

What can I ask now that I never could before?





Lunch Break

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